

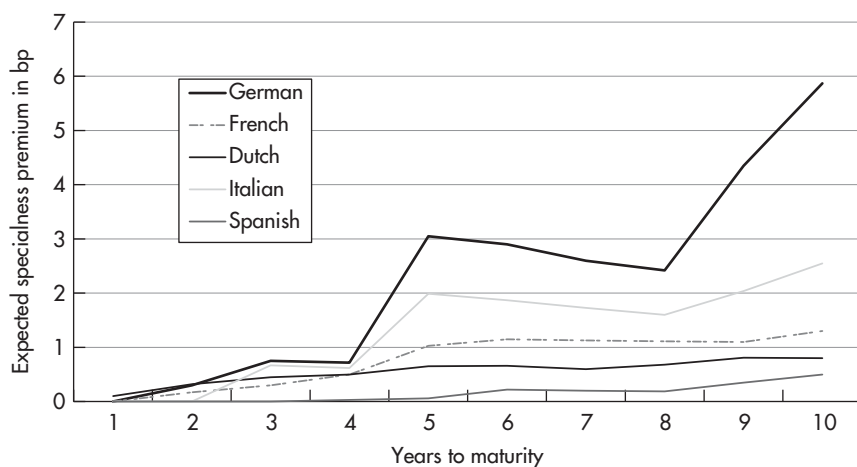


FIGURE 12.2 Historical average for the GC–repo basis swap for different EUR sovereign issues.

Source: ABN Amro, reproduced with permission from RBS.

(i.e. the expected specialness premium) can be calculated by using historical repo specialness data and assessing how special a bond of a certain issuer and with a certain maturity is expected to become over its life. The result, based on ABN Amro’s database for repo rates, which was first published in our ABN Amro Research note “Many EUR sovereign default swaps are undervalued” from 17 April 2002 and is reproduced here with kind permission from RBS, is depicted in Figure 12.2. The effect of the Bobl and Bund futures contracts on the expected specialness is clearly discernible. Note that the formula above uses the negative values from Figure 12.2.

For a bond without any periods of (expected) specialness during its remaining life, its funding rate (repo) is always GC, and thus the fair value of the funding component of its SOFR–ASW is zero. This is a consequence of SOFR being by design close to the funding rate of government bonds: Unlike LIBOR, an unsecured term rate, SOFR is a secured O/N rate, calculated from the transactions in the repo market.

The transition from LIBOR to SOFR has therefore resulted in the proximity of the funding rates for bonds and the reference rate of swaps. The funding difference involved in asset swaps and expressed as repo–SOFR basis swap in the formula above is hence limited to the specialness of specific bonds and usually rather close to zero.

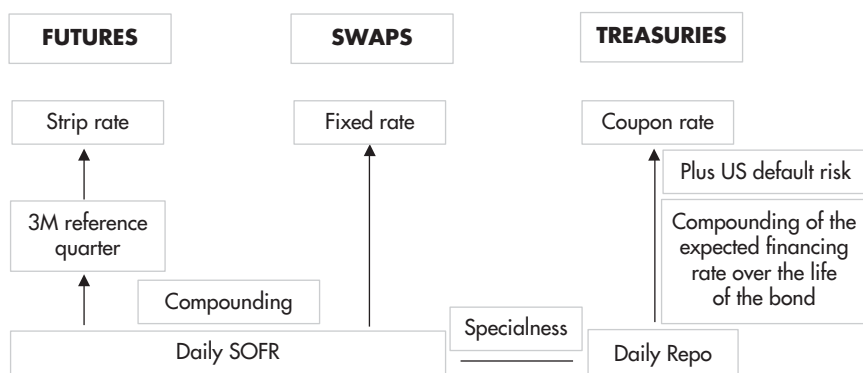


FIGURE 12.3 Integration of SOFR into a term rate in the US future, swap and bond markets.

Source: Huggins and Schaller (2022).

This observation is part of a larger picture,⁴ which can be described as convergence of the bond, swap and future markets in the US due to the transition to SOFR (Figure 12.3):

- 3M SOFR futures compound consecutive 3M forward periods of SOFR into a forward rate and hence the SOFR future strip into a term rate.
- Swaps with SOFR as floating leg exchange SOFR versus a fixed term rate.
- Treasuries can be considered as exchanging their funding rate at (a level close to) SOFR versus their coupon payments.

Hence, all three instruments, futures, swaps, and Treasuries, can be seen as (slightly different) ways to compound the same reference rate, daily SOFR values (or, in case of Treasuries, the repo rate, which is typically close to SOFR), into a term rate. Allowing for some imperfection in order to express the conceptual similarity, one could say that future strips, swaps with SOFR as reference rate and Treasuries all combine the same underlying SOFR into term rates. Abstracting from the technical differences, one might therefore consider all three instruments to be essentially the same, i.e. market prices for a certain future segment of daily SOFR.

⁴Chapter 9 of Huggins and Schaller (2022) discusses this topic in detail. The next few paragraphs contain some excerpts.

The common basis of the three markets leads to the advantage of simplifying the relationships, arbitrage, and hedging, between them. Expressing some aspects of this advantage more specifically:

- Before the financial crisis, swaps used LIBOR as reference rate, both for determining the cash flows and for discounting. Hence, the switch to overnight rates following the financial crisis and the perceived insecurity of LIBOR has resulted in dual curves. Transitioning everything to SOFR reinstates the clean and easy situation of the old days: swaps with SOFR as reference rate usually use the same curve, i.e. SOFR, both for determining the cash flows and for discounting. One consequence is that the present value of the floating side at inception is equal to 100.
- And the daily SOFR values are usually compounded for calculating cash flows and discount factors in the same manner as the settlement price of 3M future contracts. Hence, the SOFR future strip applies in principle the same calculation as swaps with SOFR as reference rate. This link allows using SOFR futures as alternative to swaps, for example, for hedging US Treasuries. One consequence is the increase in liquidity and decrease in hedging costs. In order to establish this nice principle in practice, different payment frequencies (e.g. 3M for futures and 6M for Treasuries) and daycount conventions (usually actual/360 for money market and actual/actual for Treasuries) need to be addressed. Moreover, as an OTC product, the swap counterparties can agree to all sorts of discounting curves and payment times. For example, swaps with SOFR as reference rate cleared by the CME have a 2-day payment delay. On top of all this come the issues of date mismatches between the coupon periods and the reference periods of SOFR contracts.⁵
- And as we have just seen, the migration to asset swaps with SOFR as reference rate eliminates the basis between unsecured and secured rates inherent in asset swaps with LIBOR as reference rate. Assuming a Treasury to always fund at SOFR (in particular never becoming special) and ignoring credit exposure, payment frequency and daycount differences, the transition to SOFR allows considering swaps and bonds to be conceptually the same: an exchange of an overnight secured funding rate (SOFR) into a term rate. The future, swap, and bond markets are three different places to get a price for the same collections of future SOFR values, for example, over the next two years. Another way to express this link is to say that under these assumptions, the contribution of funding differences to the asset swap spreads of Treasuries is zero.

⁵Chapter 7 of Huggins and Schaller (2022).

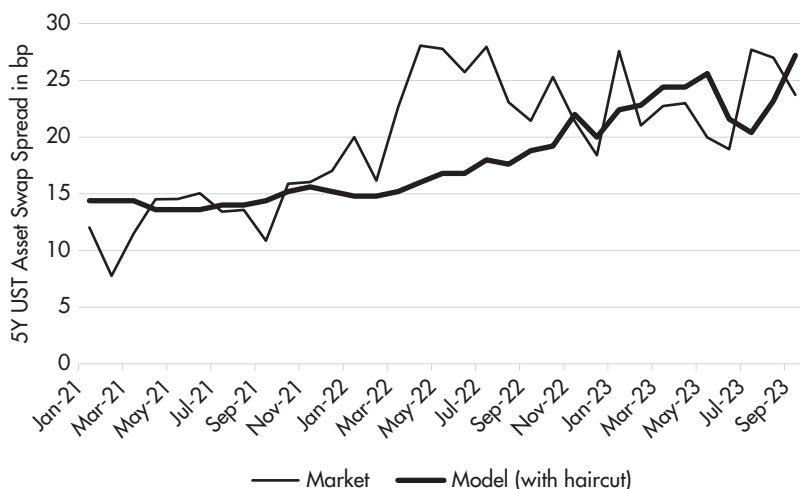


FIGURE 12.4 Actual SOFR–ASW of 5Y US Treasuries versus the model forecast.

Source: Authors.

Data period: Mid-monthly data from January 2021 to September 2023. As the CDS quotes infrequently, some interpolation assumptions were required.

Moving from theory to practice, Figure 12.4 compares the model forecast with the actual evolution of SOFR–ASW for 5Y US Treasuries. In order to minimize the structural break from the introduction of the standing repo facility (SRF) (see Chapter 11), we have chosen data points around the middle of each month, avoiding the spikes at quarter end pre-SRF.

It turns out, that without including the haircut financing costs, the model systematically underestimates the SOFR–ASW level. Hence, adding a term for haircut costs by assuming reasonable funding rates is vital to achieve the good fit to the market visible in Figure 12.4. Applying the haircut of 2% shown in Table 18.1 and a funding rate of 5% above SOFR results in 10 bp haircut costs. Actually, one could also pursue the opposite direction and estimate the haircut (financing) costs of the marginal market participant via the spread between actual and modeled SOFR–ASW. At the time of writing (September 2023), this estimate would have been 8 bp.

Once the haircut costs are added to the model value, it follows the actual market quite well. The only exception has occurred at the beginning of 2022, when the actual SOFR–ASW exceeded the model value (with haircut) by about 10 bp. Since this deviation has coincided with a strong sell-off in the Treasury market and disappeared afterwards, it seems to be an indication of a flow-related market mispricing rather than of a problem with the model.

While the empirical data for testing SOFR–ASW models is too short to draw definite conclusions, so far the model seems to reflect actual market behavior: readers are encouraged to repeat this test once more data will have become available.

Swap Spreads Versus Unsecured O/N Rates (OIS Swap Spreads)

In case of the floating rate of the asset swap being an O/N rate (we use ESTR in the following expressions, which hold true for other O/N rates as well), the asset swap spread of a bond versus ESTR⁶ (the “ESTR–ASW”) is given by the following special case of the general formula above:

$$\text{ESTR-ASW} = \text{repo-ESTR basis swap} + \text{adjusted CDS}.$$

The repo–ESTR basis swap can be thought of as consisting of two components: first, a repo–GC basis swap, second, a GC–ESTR basis swap. Seen from this perspective, it extends the repo–GC basis swap (specialness) applied in the formula for SOFR–ASW with a basis swap between secured (GC) and unsecured (ESTR) O/N rates.

As ESTR–ASW contain the funding difference between secured (repo) and unsecured (ESTR) rates, they are driven by the capital treatment discussed in Chapter 11. This is a difference to SOFR–ASW, which are independent of the cost of capital and BIS rules, explaining the usually higher level and volatility of ESTR–ASW compared to SOFR–ASW.

Under the assumption that also unsecured O/N rates involve a negligible credit risk, ESTR–ASW can be priced by using the simple formula $g \cdot q \cdot d$ for the funding difference, i.e. the fair value of the GC–ESTR basis swap, and adding both the expected specialness as explained above and the adjusted CDS level.

In contrast to the convergence of the markets for swaps and government bonds in the US described above, the transition from the unsecured funding rate LIBOR to the secured funding rate SOFR has caused a divergence of the markets for swaps and corporate bonds. Actually, hedging a corporate bond with a swap using SOFR as reference rate now involves the same unsecured versus secured basis, which was involved in hedging government bonds with swaps using LIBOR as reference rate.

Hence, if asset swaps or swap hedges which do not involve the secured–unsecured basis are desired, swaps with SOFR as reference rate can be used

⁶When combined with a USD-EUR CCBS, also Treasuries have ESTR–ASW and EURIBOR–ASW. And the other way round, basis swapped Bunds also have SOFR–ASW.

for government bonds and swaps with Fed Funds (in case of the US) as reference rate can be used for corporate bonds. Both (asset) swaps are connected through the SOFR–Fed Fund basis; as a result of the convergence shown in Figure 12.3, the SOFR–Fed Fund future spread contracts provide a market price and a cheap way to hedge this basis.

Swap Spreads Versus LIBOR

In case of the floating rate of the asset swap being LIBOR, the asset swap spread of a bond versus LIBOR (the “LIBOR–ASW”) is given by the following special case of the general formula above:

$$\text{LIBOR-ASW} = \text{repo-LIBOR basis swap} + \text{adjusted CDS.}$$

The repo–LIBOR basis swap can be thought of as consisting of three components: (1) a repo–GC basis swap, (2) a GC–ESTR basis swap, and (3) an ESTR–LIBOR basis swap. Seen from this perspective, it extends the repo–GC basis swap (specialness) applied in the formula for SOFR–ASW with a basis swap between secured (GC) and unsecured (ESTR) O/N rates *and* with a basis swap between unsecured O/N (ESTR) and term (LIBOR) rates.

In comparison with SOFR–ASW, LIBOR–ASW depends in addition to the specialness element also on the secured–unsecured spread and hence the capital considerations *and* on the O/N–term spread and hence on a non-negligible credit exposure to the money market counterparties. LIBOR–ASW should therefore be priced by using the model for the repo–LIBOR spread (including the credit element) for the funding difference, i.e. the fair value of the GC–LIBOR basis swap, before adding again both the expected specialness as explained above and the adjusted CDS level.

Taking a step back and overviewing the three special cases of the general concept, one realizes how the increasing difference between repo and the reference rate of the swap (first SOFR, then ESTR, finally LIBOR) results in the elements driving the asset swap spread to increase, with the calculation being expanded accordingly. Apart from the credit exposure of the bond issuer (and driving factors not captured in the model) involved in every type of swap spread, SOFR–ASW only depends on the expected specialness, ESTR–ASW also on the capital treatment driving the secured–unsecured basis, LIBOR–ASW also on the term risk. From this viewpoint, the evolution of reference rates summarized in Table 11.1 in the opposite direction (from LIBOR to SOFR) can be considered as removing one difference between the funding of bonds and swaps after another, with the convergence of the markets depicted in Figure 12.3 as the goal.

The increasing number of driving forces is reflected in a usually increasing volatility when moving from SOFR–ASW via ESTR–ASW to LIBOR–ASW. The variable CoE is particularly prone to sharp moves, but affects only ESTR–ASW and LIBOR–ASW, not SOFR–ASW. This is one reason why the ASW of Bunds versus ESTR and LIBOR tends to be more volatile than the ASW of US Treasuries versus SOFR shown in Figure 12.4. Chapter 16 will combine these two observations and discuss how the relative stability of SOFR–ASW of US Treasuries and of their spread to basis swapped Bunds results in most of the volatility of the ESTR–ASW and LIBOR–ASW of Bunds being usually absorbed by volatile basis swaps (and vice versa).

TERM STRUCTURE OF SWAP SPREADS

Summarizing the discussion above, Table 12.1 exhibits the driving factors of the different types of asset swap spreads.

By assessing the term structure of each of the driving factors one can predict the term structure of the asset swap spread curve:

- The credit risk of bond issuers has a term structure, which can, for example, be calculated by multiplying the yearly rating transition matrix with itself. It turns out that for issuers starting with a good rating (as is the case for the sovereign issuers considered in this book), the default risk increases over time. This is also reflected in the market price for that risk, i.e. the adjusted CDS quotes. While liquidity in the government CDS market is concentrated in 5Y, rare quotes for shorter or longer maturities are regularly lower or higher. In line with this observation, the asset swap spread curves tend to be upward-sloping. The SOFR–ASW curve of Treasuries usually starts around zero (as explained above) and then becomes more and more positive, which can (partly) be explained by the increasing credit exposure to the US: the longer the maturity, the higher

TABLE 12.1 Driving Factors of the Different Types of Asset Swap Spreads

Driving factor of	SOFR–ASW	OIS–ASW	LIBOR–ASW
Credit risk of bond (CDS)	Yes	Yes	Yes
Specialness	Yes	Yes	Yes
Capital requirements	No	Yes	Yes
Money market term (risk)	No	No	Yes

the chance for the US credit quality to deteriorate – while being stuck with the bond without the feature of credit ‘refreshes.’ The LIBOR–ASW curve of Bunds usually starts with negative values (reflecting the capital requirements) and then also increases. If and where it reaches positive values depends on the relationship between the CDS curve of Germany and the unsecured–secured basis on a given day.

- The specialness also has a term structure, as depicted in Figure 12.2. Since specialness is mostly caused by benchmark and delivery status, it usually increases with the maturity of the bond. Hence, the longer the bond, the higher its expected specialness and thus the lower (in case of LIBOR–ASW of Bunds usually more negative) the asset swap spread. Specialness is therefore a driving factor which imposes a term structure opposite to the credit exposure. For example, with an unsecured–secured basis of 16 bp (using the example given in Chapter 11) and an adjusted 10Y German CDS level of 15 bp, the ESTR–ASW of a 10Y Bund should be $-16 \text{ bp} + 15 \text{ bp} - 6 \text{ bp} = -7 \text{ bp}$, with the term “–6 bp” taken from the average specialness shown in Figure 12.2.
- Since the capital requirements are the same for any maturity (for the instruments considered here), the unsecured–secured basis has no term structure. Using the example from Chapter 11 again, for a cost of capital of 10%, that basis is –16 bp for all maturities. Only in case of expected future changes to one of the three variables, cost of equity, risk weighting, and core capital ratio, would there be a term structure to the basis as well. For instance, if a change to the cost of capital or to the BIS rules was expected to occur in two years, the unsecured–secured basis would exhibit a term structure from that point in time onwards.⁷
- The term (risk) involved in LIBOR–ASW is the same for all bond maturities and does not contribute to the term structure of LIBOR–ASW.
- In addition, also the haircut schedules usually have a term structure, which influences all three asset swap spread curves. While the exact quantity depends on the financing costs of the haircut, as the haircuts typically increase with the maturity of the bond, this factor generally supports steeper swap spread curves. In fact, we’ll see that the exceptionally steep haircut schedule of the BoE (Table 18.2) is partly responsible for the exceptionally steep SOFR asset swap spread curve of basis swapped Gilts (Figure 17.2).

⁷At the time of writing, we have found no basis for such expectations in the BIS documents.

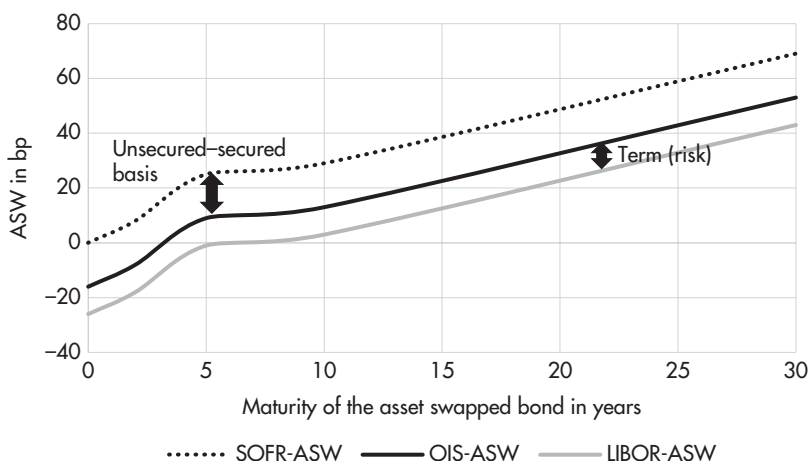


FIGURE 12.5 Schematic relationship between the three asset swap spread curves.
Source: Authors.

Cutting through Table 12.1 by columns rather than rows, we can determine the term structure of the different asset swap spread curves and find that all driving factors with a term structure are already reflected in the SOFR asset swap spread curve. Hence, all three swap spread curves are usually parallel shifts of the same slope and curvature (Figure 12.5). In more detail:

- The SOFR asset swap spread curve depends on the term structure of the CDS, specialness, and haircut curves. It typically starts at zero and is upward-sloping due to the increasing default risk and haircuts, which is somewhat mitigated by the expected specialness also increasing with the maturity.
- The OIS asset swap spread curve additionally depends on the unsecured-secured basis. In the absence of expected future changes to the cost of capital or BIS rules, this basis has no term structure. Hence, the OIS asset swap spread curve typically has the same shape as the SOFR asset swap spread curve, just starting from a lower (negative) level as given by the basis.
- The LIBOR asset swap spread curve additionally depends on the term (risk), which also has no term structure. It therefore exhibits typically the same shape as the other two, just starting from an even lower (negative) level, which also reflects the term (risk).

CYCLICALITY OF SWAP SPREADS

In the same way that we have just screened the influencing factors of swap spreads listed in Table 12.1 for term structures, we can now assess their cyclicity⁸:

- The credit risk of bond issuers exhibits a clear cyclicity: in a recession, tax revenue decreases and social benefit payments increase, negatively affecting the credit quality of a state. This is a major driving force behind the cyclicity of swap spreads, which explains their tendency to increase in times of a recession. Using the terminology from the note and the fact that recessions are typically accompanied by decreasing interest rates,⁹ one could therefore call swap spreads (with the convention Reference Rate + X) “counter-cyclical.” The next section outlines that the impact of many cyclical and counter-cyclical variables, such as bond issuance and unemployment, on swap spreads can be captured in our model via the credit component.
- We see no theoretical or empirical reason to assume cyclicity of repo specialness.
- If a recession is accompanied by a banking crisis, it is conceivable that it leads to (speculation about) more restrictive BIS rules, such as higher core capital ratios. This effect would result in lower swap spreads, i.e. be cyclical, and hence counteract the counter-cyclical impact of the credit risk of bond issuers. By contrast, if the recession leads to a sovereign downgrade, the bond may lose its 0% risk weighting, which would result in higher swap spreads and hence reinforce the counter-cyclical impact of the credit risk of bond issuers. While these events could have a material impact on swap spreads, they have been historically rare.
- As discussed at the end of Chapter 11, the difference between LIBOR and repo is an increasing function of the level of interest rates and of the probability of default of LIBOR panel banks (as well as the recovery ratio). In case of a recession leading to lower rates, for example, due to monetary stimulus, LIBOR–ASW can thus be expected to increase since the funding

⁸The term “cyclicity” refers to any correlation (positive or negative) of a time series to macro-economic cycles. Positive correlation of a time series to the level of interest rates is called “cyclical,” negative correlation “counter-cyclical.”

⁹There are exceptions such as stagflations. Actually, the current economic downturn coincides with rising interest rates. We therefore prefer to speak of the behavior of swap spreads in a “recession” rather than in a “low rate environment.”